

Ceremonial Entheogen Use in Puerto Rico — Descriptive Analysis Report

Descriptive study (unweighted); no causal or population-prevalence claims. Subgroup cells with $n < 5$ are suppressed.

1. Sample and framing

- Rows collected: **100** (target N=100 met).
- Participants (`is_participant='si'`): **72**; facilitators (`is_facilitator='si'`): **8**; dual-role: **7**.
- Outcome `ceremony_good` decoded `_0 ... _6` → 0–6, treated as **ordinal**.

2. Methods (summary)

Nonparametric throughout (Mann–Whitney U , Kruskal–Wallis H) with rank-biserial r_{rb} / epsilon-squared ε^2 effect sizes; `pna` / `not_sure` retained as categories; $n < 5$ suppression; Wilson 95% CIs for prevalences. Full rationale in `ANALYSIS_PLAN.md`.

3. Univariate description (Objectives 1–2)

Descriptive sample: **N = 87** eligible adults who reached the role fork (excludes 13 abandoned-blank submissions). This includes 14 `no/no` respondents who report no ceremony participation (retained per protocol to describe who screens out); role-specific analyses in §4–§5 restrict to participants/facilitators as noted.

3.1 Role composition

role	n	% of 100
participant_only	65	65%
screenout_no_no	14	14%
abandoned_blank	13	13%
dual	7	7%
facilitator_only	<5	1%

Facilitator and participant roles **overlap** (7 dual-role); they are not a partition. Dual-role respondents appear in both role-specific analyses below, per protocol.

3.2 Sociodemographics (analytic sample)

- **Age** (n=86): mean $\bar{x} = 42.4$, median $\tilde{x} = 41$, SD $s = 12.2$, range 21–75, IQR 34–50.

Gender identity (answered base n=85):

category	n	% answered
Woman	41	48%
Man	38	45%
Non-binary	<5	5%
I'd prefer not to answer	<5	2%
(structural skip)	<5	

Education (answered base n=86):

category	n	% answered
Bachelor's	41	48%
Masters	22	26%
Associate's Degree	10	12%
High school Diploma/GED	7	8%
I'd prefer not to answer	<5	3%
Doctorate	<5	3%
(structural skip)	<5	

Economic situation (answered base n=86):

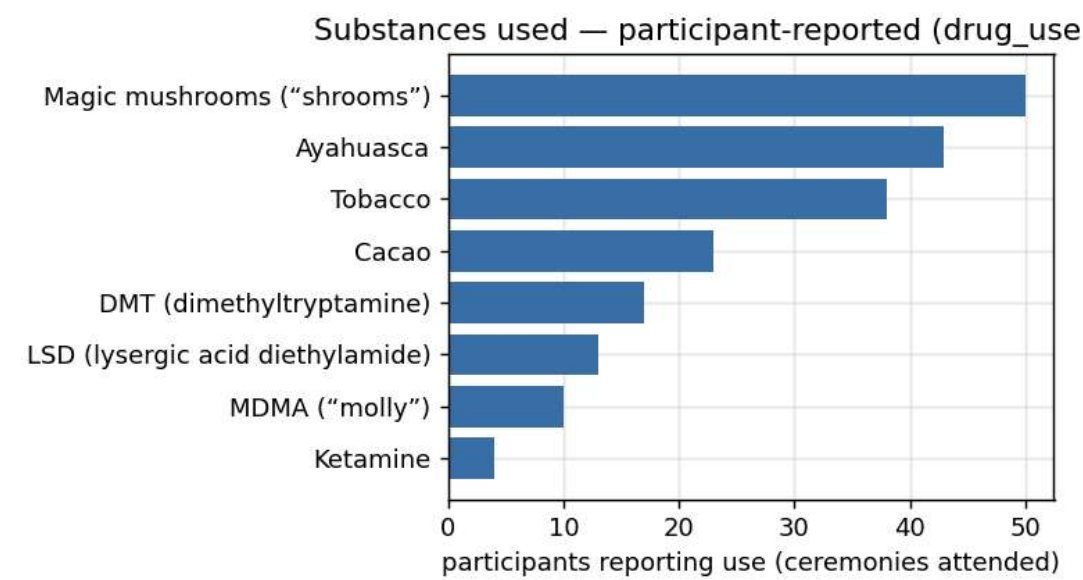
category	n	% answered
Comfortable: I can cover my needs and some extras	49	57%
b. Tight: I can cover basic needs with some difficulty	26	30%
c. Difficult: I often cannot cover my basic needs	8	9%
d. Prefer not to answer	<5	3%
(structural skip)	<5	

3.3 Residency / diaspora check

country of residence	n	% answered
Puerto Rico	82	95%
United States	<5	3%
Spain	<5	1%
(structural skip)	<5	

Residency is treated as a **descriptive covariate**, not an inclusion filter, per protocol. Non-PR respondents temper any PR-population read of the sample.

3.4 Substances (participant-reported)



The three drug variables are kept distinct: `drugs_used_facil` (facilitator-led), `drug_used_part` (participant, ceremonies attended, shown above), `drugs_used_part` (participant, most-recent ceremony).

3.5 Facilitator practice context

Facilitator-reported only (`is_facilitator='si'` , base n=8) — do not conflate with the participant-side context in §3.6. At this base, most category cells fall at or below the $n < 5$ suppression floor; tables are reported for completeness with cells masked, and should be read as **descriptive only**.

Training / experience preparing to facilitate (`training_facilitator` , select multiple; % of facilitator base):

training/experience	n	% of base
Personal experience and self-study	6	75%
Mentorship or apprenticeship learning	<5	50%
Family or ancestral tradition	<5	38%
Formal Certifications	0	0%
Academic courses or workshops	0	0%
Other	0	0%
None of the above	0	0%
Prefer not to answer	0	0%
(base n=8)		

Gives participants preparation instructions (`inst_ceremonia`) (answered base n=7):

response	n	% answered
Yes	6	86%
Prefer not to answer	<5	14%
(structural skip)	<5	

How dose is decided per participant (`dosis_choice` , answered base n=7):

dosing approach	n	% answered
I allow participants to decide their dose	<5	43%
Other	<5	29%
A conservative initial dose followed by stronger additional doses	<5	14%
A single dose for everyone	<5	14%
(structural skip)	<5	

Typical ceremony environment (ceremony_env , answered base n=7):

environment	n	% answered
3. In an outdoor space or near nature	6	86%
1. In my home	<5	14%
(structural skip)	<5	

3.6 Participant ceremony frequency and most-recent-ceremony context

Participant-reported only (is_participant='si' , base n=72).

Ceremony frequency

- **Lifetime ceremonies attended** (amount_cer) (n=67): median $\tilde{x} = 6$, mean $\bar{x} = 15.1$, SD $s = 35.7$, range 1–250, IQR 3–12.
- **Ceremonies attended in the past 12 months** (amount_cer_year) (n=67): median $\tilde{x} = 1$, mean $\bar{x} = 1.3$, SD $s = 1.7$, range 0–8, IQR 0–2.

Most-recent-ceremony context

Time since last ceremony (last_ceremony_time) (answered base n=66):

category	n	% answered
4. More than 12 months ago	30	45%
2. 2 to 6 months ago	20	30%
3. 7 to 12 months ago	11	17%
1. During the past month	<5	6%
5. Prefer not to answer	<5	2%
(structural skip)	6	

Place of last ceremony (last_ceremony_place) (answered base n=67):

category	n	% answered
3. A natural setting (e.g., forest, beach, mountain)	37	55%
5. A private home (mine or someone else's)	15	22%
2. A retreat center or wellness facility	9	13%
6. Other	<5	4%
4. An urban or residential space used informally (e.g., house, apartment, studio, clinic)	<5	3%
7. Prefer not to answer	<5	1%
(structural skip)	5	

Retreat/series vs. single ceremony (ceremony_format) (answered base n=65):

category	n	% answered
2. No, a single ceremony	48	74%
1. Yes, a retreat or a series of ceremonies	13	20%
4. Prefer not to answer	<5	5%
3. I'm not sure	<5	2%
(structural skip)	7	

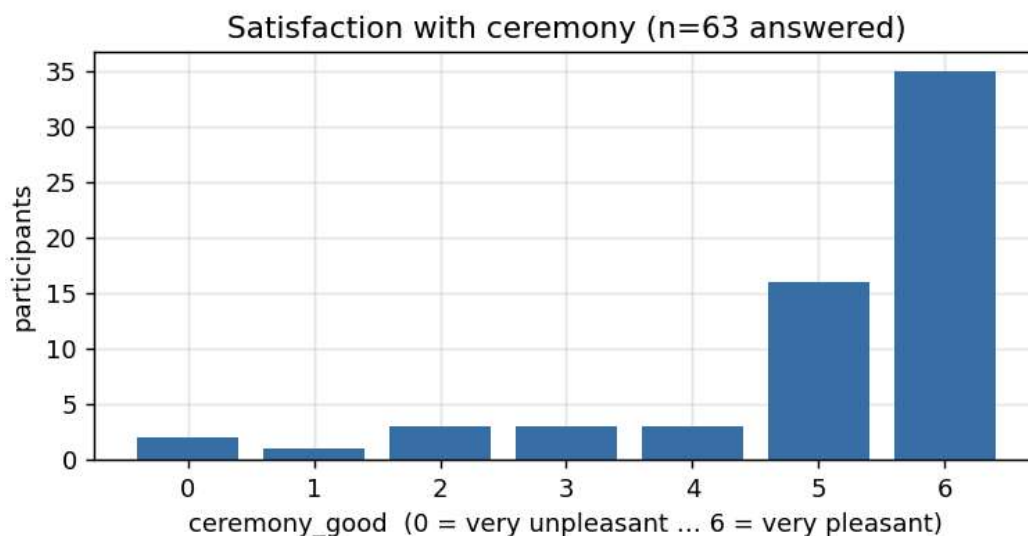
Total duration (ceremony_duration) (answered base n=65):

category	n	% answered
3. 4 to 6 hours	19	29%
5. Overnight (~12+ hours)	19	29%
4. 7 to 12 hours	14	22%
2. 1 to 3 hours	7	11%
6. 2 to 3 days	<5	6%
7. 4 to 6 days	<5	2%
1. Less than 1 hour	<5	2%
(structural skip)	7	

4. Bivariate analyses (Objectives 3–4, revised)

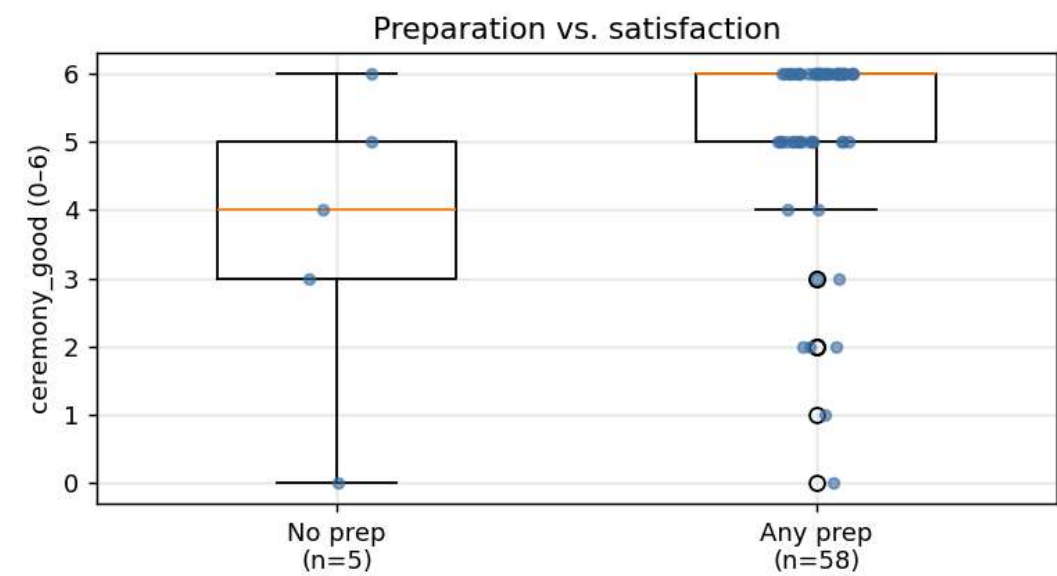
The original study proposal specified a four-arm preparation×integration design; this is **not estimable** from the collected data, as the instrument contains no clean participant *integration-received* item and the preparation split is badly unbalanced. Per protocol, we analyze **preparation** → **outcome** directly. The outcome *ceremony_good* is treated as **ordinal** (heavy ceiling), so tests are nonparametric.

4.1 Distribution and group medians



Among participants, 63/72 answered *ceremony_good* ; the remaining are item nonresponse (structural skips affect non-participants, excluded here). Distribution is strongly ceiling-weighted (median $\tilde{x} = 6$).

group	n (answered outcome)	median	IQR
Any preparation	58	6	5–6
No preparation	5	4	3–5



4.2 Primary comparison — Mann–Whitney U

Any-preparation ($n_1 = 58$) vs. no-preparation ($n_2 = 5$): $U = 218.0$, $p = 0.041$, rank-biserial $r_{rb} = 0.50$, common-language effect size CLES = 0.75.

Interpretation. The no-preparation group is $n_2 = 5$ (at the $n < 5$ suppression floor). Even a sizable effect here is statistically fragile; report as **suggestive, exploratory**, not confirmatory. Direction: preparation is associated with higher reported satisfaction.

4.3 Sensitivity — 3-level preparation (Kruskal–Wallis H)

Across prep_part levels (si: 52, a_veces: 6, no: 5): $H = 4.24$, $p = 0.120$, $\epsilon^2 = 0.04$. The no and a_veces cells are below $n < 5$ once the outcome is non-missing — interpret as descriptive only.

pn_a / not_sure are retained as substantive categories in all frequency tables but are excluded from the ordinal numeric tests (they carry no rank position).

4.4 Perceived safety by facilitator training/protocol (participant-perceived proxy; secondary, exploratory)

The training_facilitator / safe_ceremony / emergency_plan / protocol_bad_exp items as specified in the original proposal are **facilitator self-report** ($n = 8$), a different, unlinked set of respondents from the participants who report perceived safety — there is no ceremony/facilitator ID connecting a given participant to the facilitator who ran their ceremony, so those items cannot be crossed with participant-reported safety without falsely pairing unrelated respondents (the same unlinked-respondent limitation that ruled out the integration-proxy comparison in §4). This analysis instead uses the two **participant-side proxies** for facilitator training/protocol: facil_background (did the participant know about the facilitator’s background/training beforehand) and screening_quest (did the facilitator ask screening questions before the ceremony). Facilitator self-report is presented separately, descriptively, in §4.5.

4.4.1. In-ceremony comfort/safety (comfort_safe) by facil_background

group	n (answered outcome)	median	IQR
1. Yes	54	3	3–3
3. A little	8	3	2–3

group	n (answered outcome)	median	IQR
2. No	<5	2	2–2

$H = 2.24, p = 0.326, \epsilon^2 = 0.00$. Cells below $n < 5$ (no) make this descriptive/exploratory only.

4.4.2. In-ceremony comfort/safety (comfort_safe) by screening_quest

group	n (answered outcome)	median	IQR
1. Yes	39	3	3–3
2. No	15	3	2–3
3. I'm not sure	7	3	3–3

$H = 2.40, p = 0.301, \epsilon^2 = 0.01$.

4.4.3. Trust in facilitator (trust_level) by facil_background

group	n (answered outcome)	median	IQR
1. Yes	54	4	3–4
3. A little	8	3	2–3
2. No	<5	3	2–4

$H = 8.15, p = 0.017, \epsilon^2 = 0.10$. Cells below $n < 5$ (no) make this descriptive/exploratory only.

4.4.4. Trust in facilitator (trust_level) by screening_quest

group	n (answered outcome)	median	IQR
1. Yes	39	4	3–4
2. No	15	4	3–4
3. I'm not sure	7	4	3–4

$H = 0.77, p = 0.679, \epsilon^2 = -0.02$.

4.5 Facilitator self-reported training/protocol (context, descriptive only, $n = 8$)

Reported for context; **not** crossed with any safety outcome (see §4.4 for why). Every subgroup here is at or below the $n < 5$ suppression floor, so no group comparison is meaningful — counts are shown against the full facilitator base.

Training/experience (training_facilitator , select-multiple):

form of training	n	% of facilitators
Personal experience and self-study	6	75%
Mentorship or apprenticeship learning	<5	50%
Family or ancestral tradition	<5	38%
(base $n = 8$)		

Screening/safety approach (safe_ceremony , select-multiple):

approach	n	% of facilitators
I have an initial conversation or interview with the person.	6	75%

approach	n	% of facilitators
I ask about any relevant medical history or physical health condition.	<5	50%
I ask whether they are currently taking any medication.	<5	50%
I ask questions to identify any psychological or mental health challenges that could make the ceremony unsafe.	<5	38%
I welcome anyone who feels called to participate.	<5	25%
Other	<5	25%
(base $n = 8$)		

Response to a difficult experience (protocol_bad_exp , select-multiple):

response	n	% of facilitators
Singing, playing music, or using sound to guide them	5	62%
Use grounding exercises (e.g., breathing, sensory focus)	<5	50%
Offer calming or supportive words	<5	38%
Use physical touch (with prior consent or as culturally appropriate)	<5	25%
Other	<5	25%
Use a “trip killer” or something to stop the psychoactive effect (e.g., a benzodiazepine or an antipsychotic medication)	<5	12%
I have never been in this type of situation	<5	12%
(base $n = 8$)		

Emergency plan (emergency_plan):

response	n	% answered
I have an informal or unwritten approach	<5	57%
Depends on the situation	<5	43%
(structural skip)	<5	

Composite harm-reduction/readiness score (training + safety-screening approach + distress-response protocol + emergency plan, each 0/1, emergency_plan informal/depende given 0.5 partial credit; range 0–4, complete cases only):

score	n	% answered
3.5	6	86%
2.5	<5	14%
(complete-case $n = 7$)		

training_facilitator and safe_ceremony (any substantive response) are answered "yes" by every facilitator who answered them (no variance), and no facilitator selected the formal (sí) emergency_plan option (all informal/conditional) — so this branch is close to a constant. With that little variance and $n = 7$, neither a reliability statistic nor an association test with exp_crisis is meaningful here; reported for completeness only, for the same reason given in §5.3.2. The participant-side version of this index (§4.6) carries the actual inferential weight for this recommendation.

4.6 Harm-reduction practice index — participant-side score vs. ceremony_good

The composite specified in the original proposal ("prep + emergency_plan + facil_background + safe_ceremony ") mixes **facilitator self-report** items (emergency_plan , safe_ceremony , $n = 8$) with **participant self-report** items (prep_part , facil_background) — the same unlinked-respondent limitation noted for the integration proxy and the perceived-safety comparisons (§4.4): these are two different, unlinked sets of respondents, so one mixed score cannot be validly built. As in §4.4, the composite is instead built in two branch-specific versions: a **participant-side index** here (prep_any

+ `facil_background` + `screening_quest` , the three participant-answerable protective-practice items — `screening_quest` substitutes for the facilitator-side `safe_ceremony` / `emergency_plan` items as the participant-observable analog), and a **facilitator-side descriptive composite** reported for completeness at the end of §4.5. `pna` / `not_sure` are item-missing here (not scored 0), matching the convention already used to build `prep_any` .

Score distribution (0–3, complete-case $n = 54$ of 72 participants answering all three items):

score	n	% answered
3.0	35	65%
2.0	16	30%
1.0	<5	6%

Internal consistency. Cronbach's $\alpha = 0.14$ (95% CI -0.36–0.47, $n = 54$, 3 items). This is low — the three items likely capture **distinct** protective practices (preparation, facilitator vetting, being screened) rather than one shared latent construct, so the composite score below should be read as a simple count of practices present, not a validated scale.

Association with `ceremony_good` (Kruskal–Wallis across score levels):

score	n (answered outcome)	median	IQR
1	<5	4	2–5
2	14	5	5–6
3	34	6	5–6

$H = 2.09$, $p = 0.352$, $\epsilon^2 = 0.00$. Score 1 is below $n < 5$ once the outcome is non-missing — interpret as descriptive only.

4.7 Multiplicity correction — secondary/exploratory family (Benjamini–Hochberg FDR)

The primary comparison (§4.2, preparation vs. `ceremony_good` , pre-declared) is reported uncorrected. Every other bivariate test run so far — the 3-level preparation sensitivity (§4.3), the four perceived-safety-by-training comparisons (§4.4), and the harm-reduction practice index (§4.6) — forms one secondary/exploratory family, corrected jointly here.

test	raw p	BH q	significant at $q < .05$
Preparation (3-level) → <code>ceremony_good</code> (KW)	0.120	0.359	no
In-ceremony comfort/safety (<code>comfort_safe</code>) by <code>facil_background</code>	0.326	0.422	no
In-ceremony comfort/safety (<code>comfort_safe</code>) by <code>screening_quest</code>	0.301	0.422	no
Trust in facilitator (<code>trust_level</code>) by <code>facil_background</code>	0.017	0.102	no
Trust in facilitator (<code>trust_level</code>) by <code>screening_quest</code>	0.679	0.679	no
Harm-reduction practice score → <code>ceremony_good</code> (KW)	0.352	0.422	no

4.8 Multivariable ordinal model for `ceremony_good` (exploratory, hypothesis-generating)

§4.2–§4.7 are unadjusted pairwise tests; this model adjusts preparation for age and facilitator trust simultaneously. The full covariate set specified in the original proposal (age, gender, prep, trust, screening) is not estimable here: `gender_idt` has cells too thin for a 4-level categorical predictor (non-binary $n = 4$, `pna` $n = 1$), and `screening_quest` overlaps substantively with `trust_level` / `facil_background` already in the model (§4.4) — adding both risks collinearity without enough N to resolve it. Per the analysis plan's prespecified fallback, we fit a **parsimonious 3-predictor** proportional-odds model instead: `age` (years), `prep_any` (any preparation vs. none), and `trust_level_num` (trust in the facilitator, 0–4) — the two exposures already flagged as relevant in §4.2/§4.4, adjusted for baseline age.

The raw 7-level `ceremony_good` has cells as thin as $n = 1$ –3 at the low end (severe ceiling — see §4.1), which risks quasi-complete separation in an ordinal logit at this N . The outcome is therefore **collapsed to 3 ordered levels** for this model only (`ceremony_good_3lv1` ; order preserved, all univariate tables elsewhere in this report keep the full 0–6 scale).

Complete-case model sample: $n = 62$ of 63 participants with a non-missing outcome (dropped for missing `age` , `trust_level` , or `prep_part`). Outcome distribution in the model sample: low/mixed (0–3) (9), mostly positive (4–5) (19), very positive (6) (34).

predictor	OR	95% CI	p
age	0.99	0.94–1.03	0.527
prep_any (any vs. none)	5.21	0.84–32.41	0.077
trust_level	2.20	1.06–4.53	0.033

Omnibus likelihood-ratio test vs. an intercept/cutpoint-only model: $\chi^2(3) = 8.68, p = 0.034$; McFadden pseudo- $R^2 = 0.07$.

Power caveat. $n = 62$ complete cases for 5 estimated parameters (3 slopes + 2 cutpoints) — roughly 12 cases per parameter, below conventional rules of thumb for stable ordinal-regression estimates. The proportional-odds (parallel-lines) assumption is **not formally tested** here (sample too small to support it reliably) and is simply assumed. Treat every estimate in this table as **exploratory / hypothesis-generating**, not confirmatory, and do not present standalone in the manuscript without this caveat.

4.9 Substance-stratified satisfaction and safety

Stratifies by the participant's most-recent-ceremony substance (`drugs_used_part` ; full multi-ceremony substance frequencies are in §3.4). Individually reportable groups: **ayahuasca** ($n = 23$) and **magic mushrooms** ($n = 31$); every other substance — cacao, DMT, tobacco, LSD, "other" — is individually below $n < 5$ and is pooled into one **other/rare substances** group ($n = 11$), per the analysis plan's prespecified rule to combine rare substances. `pna` ($n = 1$) and non-response are excluded from this grouping variable as missing exposure (same convention as `prep_31v1`). Both tests below join one small secondary/exploratory family (§4.9.3), corrected separately from §4.7/§5.4 because the exposure (substance) cuts across both outcome domains.

4.9.1 Satisfaction (`ceremony_good`) by substance

substance group	n (answered outcome)	median	IQR
Ayahuasca	22	5	4–6
Magic mushrooms	30	6	5–6
Other/rare (combined)	11	6	4–6

$H = 4.98, p = 0.083, \epsilon^2 = 0.05$.

4.9.2 Safety (`non_con_contact`) by substance

	Ayahuasca	Magic mushrooms	Other/rare (combined)	total
1. Yes	<5	<5	0	5
2. No	22	26	11	59
3. I'm not sure	0	<5	0	<5
total	23	31	11	65
(pairwise-complete $n = 65$)				

Permutation $\chi^2 = 3.70, p = 0.486$ (10,000 label permutations, seed 20260715), Cramér's $V = 0.17$.

Affirmative (`s1`) cells are thin at this base (§5.1 already reports the pooled prevalence); read this cross-tab as **hypothesis-generating only**, same caveat as §5.3.

4.9.3 Multiplicity correction (substance-stratified family)

test	raw p	BH q	significant at $q < .05$
ceremony_good by substance group (KW)	0.083	0.166	no

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test	raw <i>p</i>	BH <i>q</i>	significant at <i>q</i> < .05
non_con_contact by substance group (perm. χ^2)	0.486	0.486	no

5. Safety / adverse-event domain (recommended; absent from proposal)

Two distinct reporters — do not merge:

- non_con_contact : **participant-reported** non-consensual physical contact (answered base ≈67).
- exp_crisis : **facilitator-reported** crisis signs *witnessed* while leading ceremonies (answered base ≈7–8) — almost entirely below the *n* < 5 floor, so reported in aggregate only.

5.1 Non-consensual contact (participant-reported)

response	n	% answered
2. No	61	91%
1. Yes	5	7%
3. I'm not sure	<5	1%
(structural skip)	5	

Prevalence = 7.5% (95% CI 3.2–16.3; *k* = 5, *n* = 67).

5.2 Facilitator-witnessed crisis signs (exp_crisis)

Facilitator base *n* = 7 answered exp_crisis . Individual crisis sub-items (including suicide_idea , panic_atk) are each below *n* < 5 and are **suppressed**; they are flagged here only to establish that serious crisis events were witnessed and are within scope. With *n* ≈ 7 facilitators, no stable rate is estimable — this motivates targeted recruitment of facilitators in future waves.

5.3 Association with facilitator training / screening / protocol (secondary, exploratory)

Disclosure note. Under the general reporting policy for this study, sensitive items such as non_con_contact and exp_crisis are otherwise reported aggregate-only, with no cross-tabulation that could re-identify a respondent. For this specific association question, the research team elected to publish cross-tabs here under the **same standard *n* < 5 cell-masking convention used for every other thin-cell table in this report** (§4.4/§4.5), rather than withholding them entirely. Even masked, several cells below are close to case-level given how few affirmative/uncertain reports exist (non_con_contact 'sí'+no estoy seguro(a)' totals *n* = 6 of 67 answered; the exp_crisis facilitator base is *n* ≈ 8). Treat every result in this subsection as **hypothesis-generating only** — not adequately powered to confirm or rule out an association.

5.3.1 Participant-reported non-consensual contact vs. participant-side training/screening proxies

Same structural-gap rationale as §4.4: facilitator self-report items are unlinked to a specific participant's ceremony, so only the participant-answered proxies (facil_background , screening_quest) can be validly crossed with a participant's own reported safety outcome. not_sure / pna are kept as their own row/column (uniform policy), not merged into si / no .

5.3.1a. non_con_contact × facil_background

	1. Yes	3. A little	2. No	4. Prefer not to answer	total
1. Yes	<5	0	<5	0	5
2. No	50	8	<5	<5	61
3. I'm not sure	0	0	<5	0	<5
total	54	8	<5	<5	67
(pairwise-complete <i>n</i> = 67)					

5.3.1b. non_con_contact × screening_quest

	1. Yes	2. No	3. I'm not sure	4. Prefer not to answer	total
1. Yes	<5	<5	0	0	5
2. No	35	14	7	<5	60
3. I'm not sure	0	0	0	<5	<5
total	39	15	7	5	66
(pairwise-complete $n = 66$)					

Permutation $\chi^2 = 13.64$, $p = 0.094$ (10,000 label permutations, seed 20260715), Cramér's $V = 0.32$.

5.3.2 Facilitator-witnessed crisis signs vs. facilitator-reported training/protocol (same respondents, $n \approx 8$)

These items sit on the same side of the survey (facilitator self-report), so — unlike §5.3.1 — there is no participant-linkage gap here; the limiting factor is purely statistical power. Select-multiple predictors are collapsed to a binary **any substantive response** indicator; emergency_plan is used at its observed categories directly.

exp_crisis (any) × training_facilitator (any substantive response)

Insufficient non-missing / non-degenerate data for a test.

exp_crisis (any) × safe_ceremony (any substantive response)

Insufficient non-missing / non-degenerate data for a test.

exp_crisis (any) × protocol_bad_exp (any substantive response)

	yes	no	total
≥1 crisis sign reported	6	0	6
no crisis sign reported	0	<5	<5
total	6	<5	7
($n = 7$)			

Fisher's exact test: odds ratio = ∞ , $p = 0.143$.

exp_crisis (any) × emergency_plan

	I have an informal or unwritten approach	Depends on the situation	total
≥1 crisis sign reported	<5	<5	6
no crisis sign reported	0	<5	<5
total	<5	<5	7
($n = 7$)			

Fisher's exact test: odds ratio = ∞ , $p = 0.429$.

Interpretation. With a facilitator base of $n \approx 8$, every 2×2 table above has at least one margin ≤ 2 ; Fisher's exact test is mathematically valid at any n , but statistical power is essentially nil. These results are reported for completeness and to establish the analytic approach for a future, larger facilitator sample — they should not be read as evidence for or against an association.

5.3.3 Harm-reduction practice index (§4.6) vs. non_con_contact

Same participant-side index built in §4.6 (`prep_any` + `facil_background` + `screening_quest` , 0–3). The facilitator-side composite (§4.5) is not tested against `exp_crisis` here — as noted there, it is close to constant at $n \approx 7$, so an association test would be statistically vacuous. The participant-side score is binarized (all three practices present vs. one or two) because the middle score cell is thin (§4.6) and `non_con_contact`'s affirmative cell is already small; a 4×3 table would leave almost every cell below $n < 5$.

	all 3 practices	1–2 practices	total
1. Yes	<5	<5	5
2. No	31	18	49
total	35	19	54
(pairwise-complete $n = 54$)			

Permutation $\chi^2 = 0.56$, $p = 0.641$ (10,000 label permutations, seed 20260715), Cramér's $V = 0.10$.

5.4 Multiplicity correction — safety-domain secondary family (Benjamini–Hochberg FDR)

The tests in §5.3 (3 participant-side + up to 4 facilitator-side) form their own secondary/exploratory family, distinct from the bivariate family in §4.7 (different outcome domain, same correction principle).

test	raw p	BH q	significant at $q < .05$
<code>non_con_contact</code> × <code>facil_background</code>	0.047	0.236	no
<code>non_con_contact</code> × <code>screening_quest</code>	0.094	0.236	no
<code>exp_crisis</code> × <code>protocol_bad_exp</code> (any)	0.143	0.238	no
<code>exp_crisis</code> × <code>emergency_plan</code>	0.429	0.536	no
Harm-reduction score (binary) × <code>non_con_contact</code>	0.641	0.641	no

6. Qualitative — open-text extraction and coding scaffold

37 open-text fields (authoritative list: `survey[type=='text']` , excluding the `nickname` / `password` identifiers) yield 285 non-empty answers, mixed Spanish/English. This is a coding scaffold to support the research team's manual thematic analysis — it does not automate interpretation.

Correction to the original field list. The proposal's prose names `add_share` as an open-text item. It is actually a `select_one si_no_pna` gate ("is there anything else you'd like to share?") for the real free-text field, `additional_info` . `add_share` is excluded here as categorical, not free text.

6.1 Answers per field

field	n answered
<code>exp_highlight</code>	58
<code>opinion_safe</code>	54
<code>prep_guidance_desc</code>	45
<code>screening_quest_type</code>	38
<code>self_prep_type</code>	30
<code>additional_info</code>	12
<code>fam_structure_other</code>	8
<code>tech_pos_exp</code>	7

field	n answered
protocol_prep	6
barriers	6
imp_factors	6
drugs_used_part_other	<5
motive_type_other	<5
dosis_choice_other	<5
protocol_other	<5
safe_ceremony_other	<5
exp_crisis_other	<5
reason_part_cer_other	<5
otro	<5
org_type_part_other	<5
last_ceremony_place_other	<5

16 field(s) with zero non-empty answers are omitted from the table above. Notably `no_comfort_safe` ("why did you feel uncomfortable/unsafe?") has **0** responses because its skip-logic condition (`comfort_safe='some_uncomf'` AND `comfort_safe='very_uncomf'`) requires one variable to equal two different values simultaneously — an instrument bug that makes the item structurally unreachable, not a low-response item.

6.2 Language mix

language_guess	n	% of answers
es	220	77%
en	54	19%
mixed/other	11	4%

`language_guess` is a rough heuristic (diacritics/marker-words for short strings, `langdetect` for longer ones) to characterize the corpus, not a validated per-item classification.

6.3 Seed keyword categories (codebook bootstrap)

theme	answers with ≥1 keyword hit
legality	19
medical_presence	19
facilitator_vetting	14
education	13
consent	<5

Categories are prespecified by the research team (legality, education, facilitator vetting, medical presence, consent), matched by ES/EN keyword stems. `opinion_safe` in particular shows a dominant legalization/regulation + facilitator-training/education cluster on inspection — a natural first code family for the team's codebook.

6.4 Word-frequency view

token	n answers
dieta	31
ceremonia	19
mental	19
salud	17
personas	15
más	14
medicina	14
experiencia	13
vida	13
espiritual	12
tener	11
educación	11
medicamentos	10
intención	9
estar	9

Stopwords removed (ES+EN); tokens below the $n < 5$ floor are not listed. This is a seed view for codebook development, not a substitute for reading the corpus.

6.5 Deliverables for the research team

- `outputs/tables/opentext_corpus.csv` — long format (`respondent_id` , `field` , `text` , `language_guess`), one row per non-empty answer, all fields.
- `outputs/tables/opentext_manual_coding.csv` — same rows plus empty `code_1` , `code_2` , `code_3` columns for manual coding by the team. This scaffold supports the team's thematic analysis; it does not replace it.

7. Data quality (flag, not drop)

7.1 Attention checks (per branch)

Facilitators judged on `atencion_1/2/3` ; participants on `atencion_3_2/4/5/6` (checks are branch-specific). Rows are **flagged, never dropped**, per protocol.

- Respondents with ≥ 1 answered-but-incorrect attention item: **2**.

7.2 Duplicate-responder screening

`ev_dup` was never populated (empty $\times 100$) and `_submitted_by` is empty (no IP/device), so duplicates are screened manually via `nickname + password` , per protocol.

- Rows sharing an identical `nickname+password` combination: **2**.
- Rows with a repeated `nickname`: **9**; repeated `password`: **12**.

All flagged rows are exported to `outputs/tables/data_quality_flags.csv` for research-team adjudication before results are finalized. A sensitivity re-run excluding adjudicated duplicates is recommended once the team confirms which flags are true duplicates.

8. Psychometric / reliability check

The instrument was purpose-built for this study (no validated PR entheogen-use instrument exists), so this is the first reliability evidence for it. The original proposal's language ("agreement scales, legal-knowledge, legal-worry, trust/comfort") implies multi-item Likert batteries, but inspecting `kobox1s.xlsx` shows the instrument does not actually contain any: every Likert-shaped `list_name` maps to at most 2 items, and most 2-item pairs split across the facilitator/participant branches — unlinked respondents, the same limitation noted for the integration proxy (§4) and the harm-reduction index (§4.6). Two genuine same-respondent pairs are testable; the rest are reported below as **not testable**, with the reason.

8.1 Trust in facilitator + in-ceremony comfort/safety

`trust_level` (5-level) and `comfort_safe` (4-level) are both participant-side, single-item measures of subjective in-ceremony experience, already treated as related outcomes in §4.4. With only **2 items**, Cronbach's α is a direct, monotonic transform of their pairwise correlation ($\alpha = 2r/(1 + r)$), and the item-total (item-rest) correlation for each item is identical to that pairwise correlation — not an independent number.

Spearman correlation: $\rho = 0.39, p = 0.002$ ($n = 64$ pairwise-complete).

Cronbach's $\alpha = 0.58$ (95% CI 0.31–0.74, complete-case $n = 64$). This reflects a positive association between reported trust and perceived safety, consistent with both tracking one underlying "the ceremony felt held/safe" experience — but with only 2 items this is suggestive, not a validated scale.

8.2 Perceived legal knowledge — general vs. most-recent ceremony

`info_est_legal_part` (general legal understanding, `list_name` `legal_know_scale`) and `legal_know_part` (most-recent-ceremony legal understanding, `list_name` `legal_know_scale_2`) are both participant-side items on the same construct, but use two different `list_name` option sets — the recent-ceremony version is **missing the bottom "not aware at all" category** the general version has. They cannot be run through the standard per-scale ordinal decode used elsewhere, so they are harmonized by matching option content into one shared 0–4 ordinal code; `pna` is excluded (same policy as every other ordinal numeric encoding in this report).

Spearman correlation: $\rho = 0.79, p = 0.000$ ($n = 55$ pairwise-complete).

Cronbach's $\alpha = 0.86$ (95% CI 0.75–0.92, complete-case $n = 55$). Reported for the same reason as §8.1: a real, if provisional, reliability signal rather than a placeholder result.

8.3 Candidates not testable, and why

candidate set	why not testable
<code>impact_positive</code> / <code>impact_negative</code> (perceived life impact, <code>agreement_scale_2</code>)	<code>impact_positive</code> has zero variance : all 67 respondents who answered selected "somewhat_agree" (67/67). A constant item cannot correlate with anything and alpha is undefined. Reported as a data-quality finding, not a scale.
<code>est_legal_worry</code> / <code>est_legal_worry_part</code> (<code>legal_worry_scale</code>)	Asymmetric skip logic: <code>est_legal_worry</code> is gated on <code>info_est_legal_part='pna'</code> — the opposite of what a parallel item would need — so only $n = 6$ ever reach it (near-constant response). <code>est_legal_worry_part</code> is asked broadly ($n = 57$). Not comparable/parallel items; likely an instrument logic bug, same category as the <code>no_comfort_safe</code> unreachable-item finding in §6.
<code>entheogen_legal_know</code> (facilitator legal-knowledge item)	Facilitator-branch item ($n = 7$), unlinked to any participant respondent — the same limitation noted for the integration proxy (§4) and perceived-safety analysis (§4.4). No matching facilitator-side item exists to pair it with.

9. Legal knowledge/worry correlates

The analysis plan lists `entheogen_legal_know`, `legal_know_part`, and `est_legal_worry(part)` as candidate predictors against "disclosure/screening participation and safety behavior." §8.3 already found two of those four legal-domain items unusable for any cross-tabulation: `est_legal_worry` (general) is gated on `info_est_legal_part='pna'`, reaching only a handful of respondents; `entheogen_legal_know` is a facilitator-branch item ($n \approx 8$), unlinked to any participant respondent (§9.5 repeats the exact counts). This section instead uses the three well-populated **participant-side** legal items — `info_est_legal_part_num` (general legal knowledge, harmonized 0–4, §8.2), `legal_know_part_num` (most-recent-ceremony legal knowledge, harmonized 0–4, §8.2), and `est_legal_worry_part` (most-recent-ceremony legal worry, 5-level `legal_worry_scale`, decoded 0–4) — against three participant-side outcomes: **disclosure** (`med_couns_part`: talked to a doctor/therapist/health professional about ceremony participation beforehand),

9.1 Disclosure to a health professional (med_couns_part)

General legal knowledge (info_est_legal_part) by med_couns_part

group	n (answered predictor)	median	IQR
1. Yes	7	2	2–4
2. No	50	2	1–4
3. I'm not sure	<5	4	4–4

$H = 2.71, p = 0.258, \epsilon^2 = 0.01$. Cells below $n < 5$ (not_sure) make this descriptive/exploratory only.

Most-recent-ceremony legal knowledge (legal_know_part) by med_couns_part

group	n (answered predictor)	median	IQR
1. Yes	8	4	3–4
2. No	47	3	2–4
3. I'm not sure	<5	4	4–4

$H = 3.61, p = 0.164, \epsilon^2 = 0.03$. Cells below $n < 5$ (not_sure) make this descriptive/exploratory only.

Most-recent-ceremony legal worry (est_legal_worry_part) by med_couns_part

group	n (answered predictor)	median	IQR
1. Yes	8	2	0–2
2. No	47	0	0–0
3. I'm not sure	<5	0	0–0

$H = 6.92, p = 0.031, \epsilon^2 = 0.09$. Cells below $n < 5$ (not_sure) make this descriptive/exploratory only.

9.2 Screened by facilitator (screening_quest)

General legal knowledge (info_est_legal_part) by screening_quest

group	n (answered predictor)	median	IQR
1. Yes	36	2	1–4
2. No	12	3	2–4
3. I'm not sure	7	2	1–4

$H = 1.77, p = 0.412, \epsilon^2 = -0.00$.

Most-recent-ceremony legal knowledge (legal_know_part) by screening_quest

group	n (answered predictor)	median	IQR
1. Yes	35	3	2–4
2. No	13	3	2–4

group	n (answered predictor)	median	IQR
3. I'm not sure	6	3	2–4

$H = 0.83, p = 0.659, \epsilon^2 = -0.02.$

Most-recent-ceremony legal worry (est_legal_worry_part) by screening_quest

group	n (answered predictor)	median	IQR
1. Yes	35	0	0–0
2. No	13	0	0–0
3. I'm not sure	6	0	0–0

$H = 0.18, p = 0.914, \epsilon^2 = -0.04.$

9.3 Unwanted contact (non_con_contact)

General legal knowledge (info_est_legal_part) by non_con_contact

group	n (answered predictor)	median	IQR
1. Yes	5	2	1–4
2. No	54	2	1–4
3. I'm not sure	<5	1	1–1

$H = 0.97, p = 0.617, \epsilon^2 = -0.02.$ Cells below $n < 5$ (not_sure) make this descriptive/exploratory only.

Most-recent-ceremony legal knowledge (legal_know_part) by non_con_contact

group	n (answered predictor)	median	IQR
1. Yes	5	4	3–4
2. No	51	3	2–4
3. I'm not sure	<5	1	1–1

$H = 3.61, p = 0.165, \epsilon^2 = 0.03.$ Cells below $n < 5$ (not_sure) make this descriptive/exploratory only.

Most-recent-ceremony legal worry (est_legal_worry_part) by non_con_contact

group	n (answered predictor)	median	IQR
1. Yes	5	0	0–2
2. No	51	0	0–0
3. I'm not sure	<5	0	0–0

$H = 0.99, p = 0.611, \epsilon^2 = -0.02.$ Cells below $n < 5$ (not_sure) make this descriptive/exploratory only.

9.4 Multiplicity correction (legal-knowledge/worry family)

All nine comparisons above (3 legal predictors × 3 outcomes) form one secondary/exploratory family, corrected jointly here — separate from §4.7/§4.9.3/§5.4 because the exposure (legal knowledge/worry) and these outcomes are not shared with those other families.

test	raw p	BH q	significant at $q < .05$
General legal knowledge (info_est_legal_part) by med_couns_part (KW)	0.258	0.581	no

test	raw <i>p</i>	BH <i>q</i>	significant at <i>q</i> < .05
Most-recent-ceremony legal knowledge (legal_know_part) by med_couns_part (KW)	0.164	0.495	no
Most-recent-ceremony legal worry (est_legal_worry_part) by med_couns_part (KW)	0.031	0.283	no
General legal knowledge (info_est_legal_part) by screening_quest (KW)	0.412	0.741	no
Most-recent-ceremony legal knowledge (legal_know_part) by screening_quest (KW)	0.659	0.741	no
Most-recent-ceremony legal worry (est_legal_worry_part) by screening_quest (KW)	0.914	0.914	no
General legal knowledge (info_est_legal_part) by non_con_contact (KW)	0.617	0.741	no
Most-recent-ceremony legal knowledge (legal_know_part) by non_con_contact (KW)	0.165	0.495	no
Most-recent-ceremony legal worry (est_legal_worry_part) by non_con_contact (KW)	0.611	0.741	no

9.5 Not tested

candidate	why not tested
est_legal_worry (general legal worry)	Gated on info_est_legal_part='pna' — only <i>n</i> = 6 ever reach it (§8.3); too sparse for a group comparison.
entheogen_legal_know (facilitator legal knowledge)	Facilitator-branch item (<i>n</i> = 7), unlinked to any participant respondent — same limitation as §4/§4.4/§8.3.

10. Reproducibility

Regenerate: PYTHONIOENCODING=utf-8 python analysis/report.py . Pipeline: config.py → data_prep.py → quality.py → qualitative.py → stats_helpers.py → report.py . All decodes derive from koboxls.xlsx .